

attack, the malicious actor tries to access the communication between the user and the system. Confidential information is obtained by sniffing on the network.

Denial of service attack: In this type of attack, the malicious actor prevents access to the system, and the computer or server is unable to perform its intended function. This makes the system unusable, affecting its availability.

SQL injection attack: SQL is a database and answers all the queries it receives. The attackers look at vulnerable applications and post queries to get the information from the database.

Password attack: In this type of attack, the hacker tries to guess the passwords of user accounts. There are different types of password attacks.

- **Dictionary attack:** The attacker tries to use all the words in the dictionary in an attempt to guess the password.
- **Brute force attack:** The attacker tries to guess the various password combinations possible, through trial and error.
- **Keylogger:** This attack creates records of everything that is typed on a computer or mobile keyboard.
- **Shoulder surfing:** In this type of attack, passwords are obtained by looking over the user's shoulder.
- **Distributed denial of service or DDOS attack:** Here multiple bots are used to perform denial-of-service attacks on the target. The hacker gets control of the bots and tries to attack the target through them. It is difficult to trace the source of this attack. There are different kinds of DDOS attacks such as volume based attacks, protocol attacks and application layer attacks.

Cyber security testing

Cyber security testing involves uncovering the vulnerabilities and making a risk assessment of the system for cyber attacks. The goal is to find the loopholes and weaknesses of the system.

There are different types of security testing.

Vulnerability scanning: This involves finding the weaknesses of the system through which a hacker can gain unauthorised access to the machine. Various parts of the system such as open ports are checked for vulnerabilities.

Penetration testing: A pen test attacks a system to help find the vulnerabilities in it.

Security auditing: The audit of the system is done to find the security vulnerabilities.

Security risk assessment: In this kind of testing the risks and vulnerabilities in the system are analysed. This involves a process of identifying the security loopholes in the system, analysing the vulnerabilities and addressing them. Historic problems with the system are evaluated from the attacker's point of view

Ethical hacking: Ethical hacking is a process of understanding the vulnerabilities in the system, installed applications, networking protocols used and in the databases. It tries to understand the weak points in the system and lawfully attack it to evaluate and resolve its vulnerabilities.

Scapy: The network packet manipulation tool

Now that we know a little about the risks associated with the Internet, we will have a look at Scapy, which is a packet crafting library for Python 2 and Python 3.

Scapy is a powerful packet crafting tool. It can forge or decode the packets of a wide variety of protocols, send them on the Internet, and respond to the replies. It can be used to handle the classical tasks of cyber security such as network scanning, probing, unit tests, performing attacks, network discovery and debugging of network attacks.

Scapy can do the following:

- Network scanning
- Network attack

- Automated analysis of *pcap* files from Wireshark

Cyber security activities

Port scanning: Ports are the doors to the network through which information is received and transmitted. Port scanning determines which ports are open and can receive information, and what services are available from that system.

Network fingerprinting: Network fingerprinting is the process of finding the information on the network device such as the version of the operating system, network protocols, active network services, databases installed, software applications and configuration, and so on. An attacker's first activity is to find as much information as possible such as the target platform, software application, databases and configuration, network topology and architecture. This information can be used to find vulnerabilities and plan for the attack.

Packet sniffing: In this, hardware or software is used to observe the packets flowing through the network. This could be between two computers or multiple devices on the network. It is also called packet analysis or protocol analysis. Packet sniffers work by intercepting and logging the traffic on the network. The raw packets are analysed to decode their protocol information.

Packet forging: Packet forging is the manipulation of network packets sent to the devices. This is done to take advantage of the vulnerabilities in the software. In packet forging, the protocol header or payload information is manipulated and sent to the device.

After getting the necessary information through port scanning, fingerprinting, and understanding the vulnerabilities of the system, a cyber attack is carried out to disrupt and disable a system.